See the definitions of key terms in the glossary

Create a bookmark

Switch to whiteboard view

Solution

$$\text{heat change} = m \times c \times \Delta T$$
$$= 10.0\ \text{g} \times 0.385\ \text{J g}^{-1}\ °\text{C}^{-1} \times -60.0\ °\text{C}$$ (the value is negative as the Cu has lost heat)
$$= -231\ \text{J}$$

Exercises

1 When a sample of NH_4SCN is mixed with solid $Ba(OH)_2.8H_2O$ in a glass beaker, the mixture changes to a liquid and the temperature drops sufficiently to freeze the beaker to the table. Which statement is true about the reaction?

A The process is endothermic and ΔH is –
B The process is endothermic and ΔH is +
C The process is exothermic and ΔH is –
D The process is exothermic and ΔH is +

2 Which one of the following statements is *true* of all exothermic reactions?

A They produce gases.
B They give out heat.
C They occur quickly.
D They involve combustion.

3 If 500 J of heat is added to 100.0 g samples of each of the substances below, which will have the largest temperature increase?

	Substance	Specific heat capacity / J g^{-1} K^{-1}
A	gold	0.129
B	silver	0.237
C	copper	0.385
D	water	4.18

4 The temperature of a 5.0 g sample of copper increases from 27 °C to 29 °C. Calculate how much heat has been added to the system. (Specific heat capacity of Cu = 0.385 J g^{-1} K^{-1})

A 0.770 J **B** 1.50 J **C** 3.00 J **D** 3.85 J

5 Consider the specific heat capacity of the following metals.

Metal	Specific heat capacity / J g^{-1} K^{-1}
Al	0.897
Be	1.82
Cd	0.231
Cr	0.449

1 kg samples of the metals at room temperature are heated by the same electrical heater for 10 min. Identify the metal which has the highest final temperature.

A Al **B** Be **C** Cd **D** Cr

6 The specific heat of metallic mercury is 0.138 J g^{-1} °C^{-1}. If 100.0 J of heat is added to a 100.0 g sample of mercury at 25.0 °C, what is the final temperature of the mercury?

Enthalpy changes and the direction of change

There is a natural direction for change. When we slip on a ladder, we go down, not up. The direction of change is in the direction of lower stored energy. In a similar way, we expect methane to burn when we strike a match and form carbon dioxide and water. The chemicals are changing in a way which reduces their enthalpy (Figure 5.5).

CHALLENGE YOURSELF

2 Suggest an explanation for the pattern in specific heat capacities of the metals in Exercise 3.

217

Animation
Select the icon to see a related animation

Quiz
Select the icon to take an interactive quiz to test your knowledge

Worked solutions
Select the icon at the end of the chapter to view worked solutions to exercises in this chapter

Answers
Select the icon at the end of the chapter to view answers to exercises in this chapter

01 Stoichiometric relationships

Essential ideas

1.1 Physical and chemical properties depend on the ways in which different atoms combine.

1.2 The mole makes it possible to correlate the number of particles with a mass that can be measured.

1.3 Mole ratios in chemical equations can be used to calculate reacting ratios by mass and gas volume.

The reaction between ignited powdered aluminium and iron oxide, known as the thermite reaction:

$$2Al(s) + Fe_2O_3(s) \rightarrow Al_2O_3(s) + 2Fe(s)$$

Significant heat is released by the reaction, and it is used in welding processes including underwater welding. The stoichiometry of the reaction, as shown in the balanced equation, enables chemists to determine the reacting masses of reactants and products for optimum use.

The birth of chemistry as a physical science can be traced back to the first successful attempts to quantify chemical change. Carefully devised experiments led to data that revealed one simple truth. *Chemical change involves interactions between particles that have fixed mass*. Even before knowledge was gained of the atomic nature of these particles and of the factors that determine their interactions, this discovery became the guiding principle for modern chemistry. We begin our study with a brief introduction to this particulate nature of matter, and go on to investigate some of the ways in which it can be quantified.

The term **stoichiometry** is derived from two Greek words – *stoicheion* for element and *metron* for measure. Stoichiometry describes the relationships between the amounts of reactants and products during chemical reactions. As it is known that matter is conserved during chemical change, stoichiometry is a form of book-keeping at the atomic level. It enables chemists to determine what amounts of substances they should react together and enables them to predict how much product will be obtained. The application of stoichiometry closes the gap between what is happening on the atomic scale and what can be measured.

In many ways this chapter can be considered as a toolkit for the mathematical content in much of the course. It covers the universal language of chemistry, **chemical equations**, and introduces the **mole** as the unit of amount. Applications include measurements of mass, volume, and concentration.

You may choose not to work through all of this at the start of the course, but to come back to these concepts after you have gained knowledge of some of the fundamental properties of chemical matter in Chapters 2, 3, and 4.

1.1 Introduction to the particulate nature of matter and chemical change

Understandings:

- Atoms of different elements combine in fixed ratios to form compounds, which have different properties from their component elements.

  Guidance
  Names and symbols of the elements are in the IB data booklet in Section 5.
- Mixtures contain more than one element and/or compound that are not chemically bonded together and so retain their individual properties.
- Mixtures are either homogeneous or heterogeneous.

01 Stoichiometric relationships

Applications and skills:

- Deduction of chemical equations when reactants and products are specified.

  Guidance

  Balancing of equations should include a variety of types of reactions.

- Application of the state symbols (s), (l), (g), and (aq) in equations.
- Explanation of observable changes in physical properties and temperature during changes of state.

  Guidance

  - *Names of the changes of state – melting, freezing, vaporization (evaporation and boiling), condensation, sublimation and deposition – should be covered.*
  - *The term 'latent heat' is not required.*

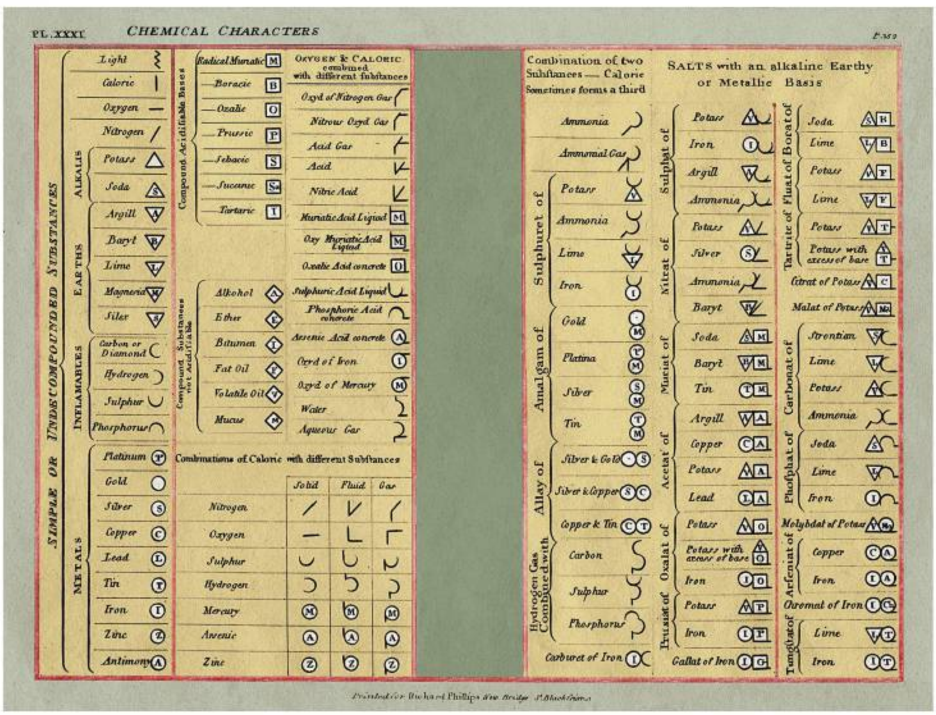
Antoine-Laurent Lavoisier, French chemist (1743–1794).

Antoine-Laurent Lavoisier (1743–1794) is often called the 'father of chemistry'. His many accomplishments include the naming of oxygen and hydrogen, the early development of the metric system, and a standardization of chemical nomenclature. Most importantly, he established an understanding of combustion as a process involving combination with oxygen from the air, and recognized that matter retains its mass through chemical change, leading to the law of conservation of mass. In addition, he compiled the first extensive list of elements in his book *Elements of Chemistry* (1789). In short, he changed chemistry from a qualitative to a quantitative science. But, as an unpopular tax collector in France during the French Revolution and Terror, he was tried for treason and guillotined in 1794. One and a half years after his death he was exonerated, and his early demise was recognized as a major loss to France.

A chemical element is a single pure substance, made of only one type of atom.

Chemical elements are the fundamental building blocks of chemistry

The English language is based on an alphabet of just 26 letters. But, as we know, combining these in different ways leads to an almost infinite number of words, and then sentences, paragraphs, books, and so on. It is similar to the situation in chemistry, where the 'letters' are the single substances known as **chemical elements**. There are



Pictographic symbols used at the beginning of the 18th century to represent chemical elements and compounds. They are similar to those of the ancient alchemists. As more elements were discovered during the 18th century, attempts to devise a chemical nomenclature led to the modern alphabetic notational system. This system was devised by the Swedish chemist Berzelius and introduced in 1814.

only about 100 of these, but because of the ways in which they combine with each other, they make up the almost countless number of different chemical substances in our world.

In Chapter 2 we will learn about atomic structure, and how each element is made up of a particular type of atom. The atoms of an element are all the same as each other (with the exception of isotopes, which we will also discuss in Chapter 2), and are different from those of other elements. It is this distinct nature of its atoms that gives each element its individual properties. A useful definition of an atom is that it is the smallest particle of an element to show the characteristic properties of that element.

To help communication in chemistry, each element is denoted by a **chemical symbol** of either one upper case letter, or one upper case letter followed by a lower case letter. A few examples are given below.

Name of element	Symbol
carbon	C
fluorine	F
potassium	K
calcium	Ca
mercury	Hg
tungsten	W

You will notice that often the letter or letters used are derived from the English name of the element, but in some cases they derive from other languages. For example, Hg for mercury comes from Latin, whereas W for tungsten has its origin in European dialects. Happily, these symbols are all accepted and used internationally, so they do not need to be translated. A complete list of the names of the elements and their symbols is given in Section 5 of the IB data booklet.

Chemical compounds are formed from more than one element

Some elements, such as nitrogen and gold, are found in **native** form, that is uncombined with other elements in nature. But more commonly, elements exist in chemical combinations with other elements, in substances known as **chemical compounds**. Compounds contain a fixed proportion of elements, and are held together by chemical bonds (discussed in Chapter 4). The bonding between atoms in compounds changes their

Assorted minerals, including elements such as sulfur and silver, and compounds such as Al_2O_3 (sapphire) and CaF_2 (fluorite). Most minerals are impure and exist as mixtures of different elements and compounds.

The International Union of Pure and Applied Chemistry (IUPAC) was formed in 1919 by chemists from industry and academia. Since then it has developed into the world authority on chemical nomenclature and terminology. It has succeeded in fostering worldwide communications in the chemical sciences and in uniting academic, industrial, and public sector chemistry in a common language.

Chemistry is a very exact subject, and it is important to be careful in distinguishing between upper and lower case letters. For example, Co (cobalt, a metallic element) means something completely different from CO (carbon monoxide, a poisonous gas).

The number of elements that exist is open to change as new ones are discovered, although there is often a time-lag between a discovery and its confirmation by IUPAC. During this time a provisional systematic three-letter symbol is used, using Latin abbreviations to represent the atomic number. The letters u (un) = 1, b (bi) = 2, t (tri) = 3 and so on are used. So the provisional element of atomic number 118 will continue to be known as ununoctium or uuo until it is confirmed and a name formally agreed according to the process established by IUPAC.

A compound is a chemical combination of different elements, containing a fixed ratio of atoms. The physical and chemical properties of a compound are different from those of its component elements.

properties, so compounds have completely different properties from those of their component elements.

A classic example of this is that sodium, Na, is a dangerously reactive metal that reacts violently with water, while chlorine, Cl_2, is a toxic gas used as a chemical weapon. Yet when these two elements combine, they form the compound sodium chloride, NaCl, a white crystalline solid that we sprinkle all over our food.

A combustion spoon holding sodium, Na, is lowered into a gas jar containing chlorine, Cl_2. The vigorous reaction produces white crystals of sodium chloride, NaCl.

$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$

The properties of the compound are completely different from those of its component elements.

Compounds are described using the chemical symbols for elements. A subscript is used to show the number of atoms of each element in a unit of the compound. Some examples are given below. (The reasons for the different ratios of elements in compounds will become clearer after we have studied atomic structure and bonding in Chapters 2 and 4.)

Name of compound	Symbol	Name of compound	Symbol
sodium chloride	NaCl	water	H_2O
potassium oxide	K_2O	glucose	$C_6H_{12}O_6$
calcium bromide	$CaBr_2$	ammonium sulfate	$(NH_4)_2SO_4$

Chemical equations summarize chemical change

The formation of compounds from elements is an example of **chemical change** and can be represented by a **chemical equation**. A chemical equation is a representation using chemical symbols of the simplest ratio of atoms, as elements or in compounds, undergoing chemical change. The left-hand side shows the **reactants** and the right-hand side the **products**.

A chemical equation shows:

direction of change

reactants ⟶ products

For example: calcium + chlorine → calcium chloride

$$Ca + Cl_2 \rightarrow CaCl_2$$

As atoms are neither created nor destroyed during a chemical reaction, *the total number of atoms of each element must be the same on both sides of the equation*. This is known as **balancing the equation**, and uses numbers called **stoichiometric coefficients** to denote the number of units of each term in the equation.

Chemical equations are the universal language of chemistry. What other languages are universal, and to what extent do they help or hinder the pursuit of knowledge?

For example: hydrogen + oxygen → water

$$2H_2 + O_2 \rightarrow 2H_2O$$

coefficients

	total on left side	total on right side
hydrogen atoms	4	4
oxygen atoms	2	2

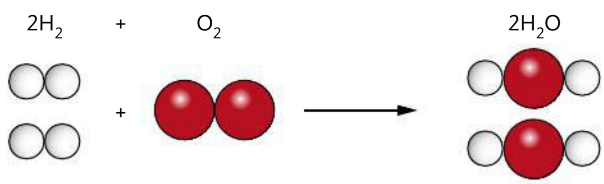


Figure 1.1 When hydrogen and oxygen react to form water, the atoms are rearranged, but the number of atoms of each element remains the same.

Note that when the coefficient is 1, this does not need to be explicitly stated.

Chemical equations are used to show all types of reactions in chemistry, including reactions of decomposition, combustion, neutralization, and so on. Examples of these are given below and you will come across very many more during this course. Learning to write equations is an important skill in chemistry, which develops quickly with practice.

Worked example

Write an equation for the reaction of thermal decomposition of sodium hydrogencarbonate ($NaHCO_3$) into sodium carbonate (Na_2CO_3), water (H_2O), and carbon dioxide (CO_2).

Solution

First write the information from the question in the form of an equation, and then check the number of atoms of each element on both sides of the equation.

$$NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$$

	total on left side	**total on right side**
sodium atoms	1	2
hydrogen atoms	1	2
carbon atoms	1	2
oxygen atoms	3	6

In order to balance this we introduce coefficient 2 on the left.

$$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$$

Finally check that it is balanced for each element.

An equation by definition has to be *balanced*, so do not expect this to be specified in a question. After you have written an equation, *always* check the numbers of atoms of each element on both sides of the equation to ensure it is correctly balanced.

When a question refers to 'heating' a reactant or to 'thermal decomposition', this does not mean the addition of oxygen, only that heat is the source of energy for the reaction. If the question refers to 'burning' or 'combustion', this indicates that oxygen is a reactant.

NATURE OF SCIENCE

Early ideas to explain chemical change in combustion and rusting included the 'phlogiston' theory. This proposed the existence of a fire-like element that was released during these processes. The theory seemed to explain some of the observations of its time, although these were purely qualitative. It could not explain later quantitative data showing that substances actually gain rather than lose mass during burning. In 1783, Lavoisier's work on oxygen confirmed that combustion and rusting involve combination with oxygen from the air, so overturning the phlogiston theory. This is a good example of how the evolution of scientific ideas, such as how chemical change occurs, is based on the need for theories that can be tested by experiment. Where results are not compatible with the theory, a new theory must be put forward, which must then be subject to the same rigour of experimental test.

Exercises

1 Write balanced chemical equations for the following reactions:

(a) The decomposition of copper carbonate ($CuCO_3$) into copper(II) oxide (CuO) and carbon dioxide (CO_2).
(b) The combustion of magnesium (Mg) in oxygen (O_2) to form magnesium oxide (MgO).
(c) The neutralization of sulfuric acid (H_2SO_4) with sodium hydroxide ($NaOH$) to form sodium sulfate (Na_2SO_4) and water (H_2O).
(d) The synthesis of ammonia (NH_3) from nitrogen (N_2) and hydrogen (H_2).
(e) The combustion of methane (CH_4) to produce carbon dioxide (CO_2) and water (H_2O).

2 Write balanced chemical equations for the following reactions:

(a) $K + H_2O \rightarrow KOH + H_2$
(b) $C_2H_5OH + O_2 \rightarrow CO_2 + H_2O$
(c) $Cl_2 + KI \rightarrow KCl + I_2$
(d) $CrO_3 \rightarrow Cr_2O_3 + O_2$
(e) $Fe_2O_3 + C \rightarrow CO + Fe$

Remember when you are balancing an equation, change the stoichiometric coefficient but never change the subscript in a chemical formula.

When balancing equations, start with the most complex species, and leave terms that involve a single element to last.

Exercises

3 Use the same processes to balance the following examples:

(a) $C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$
(b) $NH_3 + O_2 \rightarrow NO + H_2O$
(c) $Cu + HNO_3 \rightarrow Cu(NO_3)_2 + NO + H_2O$
(d) $H_2O_2 + N_2H_4 \rightarrow N_2 + H_2O + O_2$
(e) $C_2H_7N + O_2 \rightarrow CO_2 + H_2O + N_2$

A chemical equation can be used to assess the efficiency of a reaction in making a particular product. The **atom economy** is a concept used for this purpose and is defined as:

$$\%\ \text{atom economy} = \frac{\text{mass of desired product}}{\text{total mass of products}} \times 100$$

Note that this is different from % yield discussed later in this chapter, which is calculated using only one product and one reactant. Atom economy is an indication of how much of the reactants ends up in the required products, rather than in waste products. A higher atom economy indicates a more efficient and less wasteful process. The concept is increasingly used in developments in green and sustainable chemistry. This is discussed further in Chapters 12 and 13.

Mixtures form when substances combine without chemical interaction

A mixture is composed of two or more substances in which no chemical combination has occurred.

Air is described as a **mixture** of gases because the separate components – different elements and compounds – are interspersed with each other, but are not chemically combined. This means, for example, that the gases nitrogen and oxygen when mixed in air retain the same characteristic properties as when they are in the pure form. Substances burn in air because the oxygen present supports combustion, as does pure oxygen.

Tide of oily water heading towards Pensacola Beach, Florida, USA, after the explosion of the offshore drilling unit in the Gulf of Mexico in 2010. The oil forms a separate layer as it does not mix with the water.

Another characteristic of mixtures is that their composition is not fixed. For example, air that we breathe in typically contains about 20% by volume oxygen, whereas the air that we breathe out usually contains only about 16% by volume oxygen. It is still correct to call both of these mixtures of air, because there is no fixed proportion in the definition.

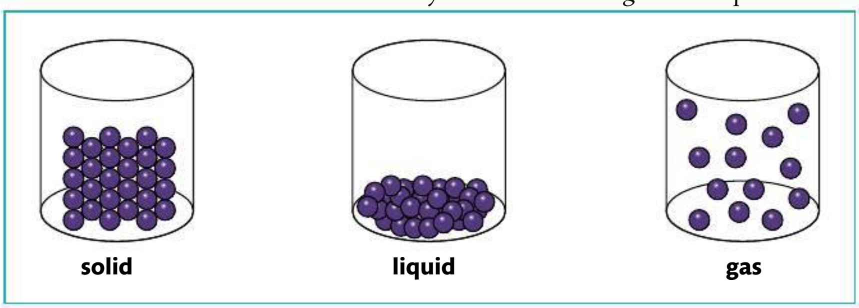
Air is an example of a **homogeneous mixture**, meaning that it has uniform composition and properties throughout. A solution of salt in water and a metal alloy such as bronze are also homogeneous. By contrast, a **heterogeneous mixture** such as water and oil has non-uniform composition, so its properties are not the same throughout. It is usually possible to see the separate components in a heterogeneous mixture but not in a homogeneous mixture.

Because the components retain their individual properties in a mixture, we can often separate them relatively easily. The technique we choose to achieve this will take advantage of a suitable difference in the physical properties of the components, as shown in the table below. Many of these are important processes in research and industry and are discussed in more detail in the following chapters.

Mixture	Difference in property of components	Technique used
sand and salt	solubility in water	solution and filtration
hydrocarbons in crude oil	boiling point	fractional distillation
iron and sulfur	magnetism	response to a magnet
pigments in food colouring	adsorbtion to solid phase	paper chromatography
different amino acids	net charge at a fixed pH	gel electrophoresis

Matter exists in different states determined by the temperature and the pressure

From our everyday experience, we know that all matter (elements, compounds, and mixtures) can exist in different forms depending on the temperature and the pressure. Liquid water changes into a solid form, such as ice, hail, or snow, as the temperature drops and it becomes a gas, steam, at high temperatures. These different forms are known as the **states of matter** and are characterized by the different energies of the particles.



increasing temperature

increasing kinetic energy of particles

- particles close packed
- inter-particle forces strong, particles vibrate in position
- fixed shape
- fixed volume

- particles more spaced
- inter-particle forces weaker, particles can slide over each other
- no fixed shape
- fixed volume

- particles spread out
- inter-particle forces negligible, particles move freely
- no fixed shape
- no fixed volume

Ocean oil spills are usually the result of accidents in the industries of oil extraction or transport. The release of significant volumes of oil causes widespread damage to the environment, especially wildlife, and can have a major impact on local industries such as fishing and tourism. Efforts to reduce the impact of the spill include the use of dispersants, which act somewhat like soap in helping to break up the oil into smaller droplets so it can mix better with water. Concern is expressed, however, that these chemicals may increase the toxicity of the oil and they might persist in the environment. The effects of an oil spill often reach countries far from the source and are the subject of complex issues in international law. With the growth in demand for offshore drilling for oil and projected increases in oil pipelines, these issues are likely to become all the more pressing.

Figure 1.2 Representation of the arrangement of the particles of the same substance in the solid, liquid, and gas states.

Depending on the chemical nature of the substance, matter may exist as atoms such as Ar(g), or as molecules such as H_2O(l), or as ions such as Na^+ and Cl^- in NaCl(aq). The term **particle** is therefore used as an inclusive term that is applied in this text to any or all of these entities of matter.

Temperature is a measure of the average kinetic energy of the particles of a substance.

This is known as the **kinetic theory** of matter. It recognizes that the average kinetic energy of the particles is directly related to the temperature of the system. The state of matter at a given temperature and pressure is determined by the strength of forces that may exist between the particles, known as **inter-particle forces**. The average kinetic energy is proportional to the temperature in Kelvin, introduced on page 37.

Worked example

Which of the following has the highest average kinetic energy?

A	He at 100 °C	B	H_2 at 200 °C
C	O_2 at 300 °C	D	H_2O at 400 °C

Solution

Answer = D. The substance at the highest temperature has the highest average kinetic energy.

Bromine liquid, $Br_2(l)$, had been placed in the lower gas jar only, and its vapour has diffused to fill both jars. Bromine vaporizes readily at room temperature and the gas colour allows the diffusion to be observed. Because gas molecules can move independently of each other and do so randomly, a gas spreads out from its source in this way.

Liquids and gases are referred to as **fluids**, which refers to their ability to flow. In the case of liquids it means that they take the shape of their container. Fluid properties are why **diffusion** occurs predominantly in these two states. Diffusion is the process by which the particles of a substance become evenly distributed, as a result of their random movements.

Kinetic energy (KE) refers to the energy associated with movement or motion. It is determined by the mass (m) and velocity or speed (v) of a substance, according to the relationship:

$$KE = \frac{1}{2} mv^2$$

As the kinetic energy of the particles of substances at the same temperature is equal, this means there is an inverse relationship between mass and velocity. This is why substances with lower mass diffuse more quickly than those with greater mass, when measured at the same temperature. This is discussed in more detail in Chapter 14.

State symbols are used to show the states of the reactants and products taking part in a reaction. These are abbreviations, which are given in brackets after each term in an equation, as shown below.

State	Symbol	Example
solid	(s)	Mg(s)
liquid	(l)	Br_2(l)
gas	(g)	N_2(g)
aqueous (dissolved in water)	(aq)	HCl(aq)

For example:

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$

Exercises

4 Classify the following mixtures as homogeneous or heterogeneous:

(a) sand and water **(b)** smoke
(c) sugar and water **(d)** salt and iron filings
(e) ethanol and water in wine **(f)** steel

Exercises

5 Write balanced equations for the following reactions and apply state symbols to all reactants and products, assuming room temperature and pressure unless stated otherwise. If you are not familiar with the aqueous solubilities of some of these substances, you may have to look them up.

(a) $KNO_3 \rightarrow KNO_2 + O_2$ (when heated, 500°C)
(b) $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + CO_2 + H_2O$
(c) $Li + H_2O \rightarrow LiOH + H_2$
(d) $Pb(NO_3)_2 + NaCl \rightarrow PbCl_2 + NaNO_3$ (all reactants are in aqueous solution)
(e) $C_3H_6 + O_2 \rightarrow CO_2 + H_2O$ (combustion reaction)

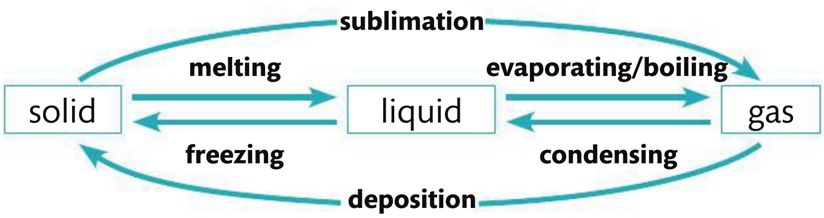
6 A mixture of two gases, X and Y, which both have strong but distinct smells, is released. From across the room the smell of X is detected more quickly than the smell of Y. What can you deduce about X and Y?

7 Ice floats on water. Comment on why this is not what you would expect from the kinetic theory of matter.

It is good practice to show state symbols in all equations, even when they are not specifically requested.

Matter changes state reversibly

As the movement or kinetic energy of the particles increases with temperature, they will overcome the inter-particle forces and change state. These state changes occur at a fixed temperature and pressure for each substance, and are given specific names shown below.



Sublimation, the direct inter-conversion of solid to gas without going through the liquid state, is characteristic at atmospheric pressure of some substances such as iodine, carbon dioxide, and ammonium chloride. Deposition, the reverse of sublimation that changes a gas directly to solid, is responsible for the formation of snow, frost, and hoar frost.

Note that evaporation involves the change of liquid to gas, but, unlike boiling, evaporation occurs only at the surface and takes place at temperatures below the boiling point.

Drying clothes. The heat of the Sun enables all the water to evaporate from the clothes.

Ice crystals, known as Hair Ice, formed by deposition on dead wood in a forest on Vancouver Island, Canada.